



Introduction

AI tools are increasingly used in CS education for code explanation, debugging, generation, and feedback. Yet most evaluations report only a single score, hiding a critical reality:

The same model can answer the same prompt differently across repeated runs.

For educational use, reliability matters as much as performance. Instructors need evidence they can trust, compare, and reproduce.

Why educators should care

Helpful AI is not enough - educational AI must be trustworthy, stable, and practical.

- Students: Inconsistent outputs confuse learners and erode trust in AI feedback.
- Instructors: Without reproducible evidence, responsible adoption is hard to justify.
- Researchers: Fair model comparisons require standardized, repeatable scaffolds.

92% of students report using AI tools but can educators trust the outputs?

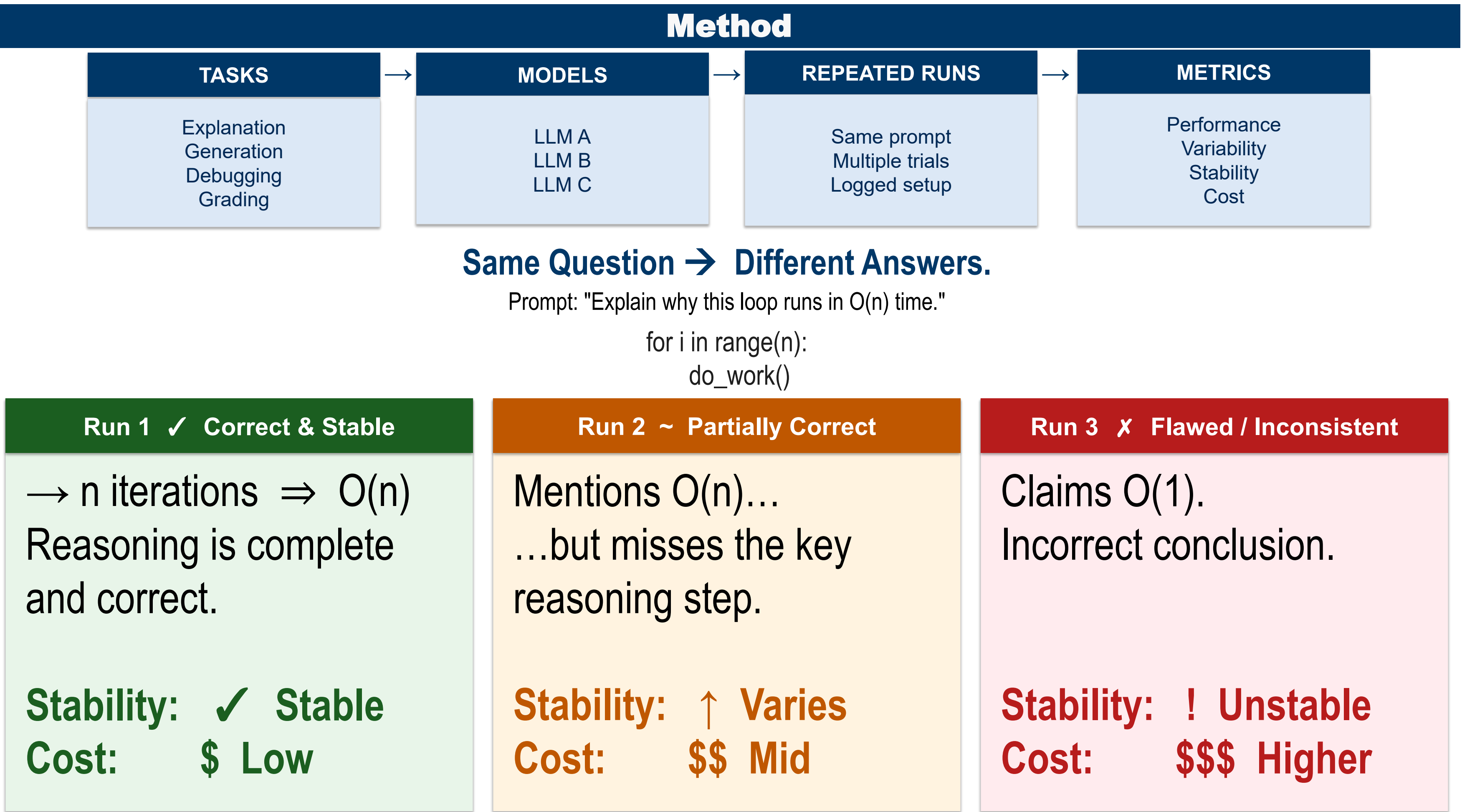
Objectives

- RQ1. How much do LLM outputs vary across repeated runs on the same CS task?
- RQ2. Which task types reveal the greatest instability?
- RQ3. Which models offer the best balance of performance, consistency, and cost?

Acknowledgements

The author gratefully acknowledges Dr. Riley Richards, Dr. Linus Yu, and the FORGE program team at Oregon Institute of Technology for their mentorship, dedicated resources, and support throughout this faculty research initiative.

Method



Repeated-run evaluation makes this variability visible and measurable - not hidden behind a single score.

Results

Early pilot evidence: accuracy alone can obscure important differences in consistency and cost across models and task types.

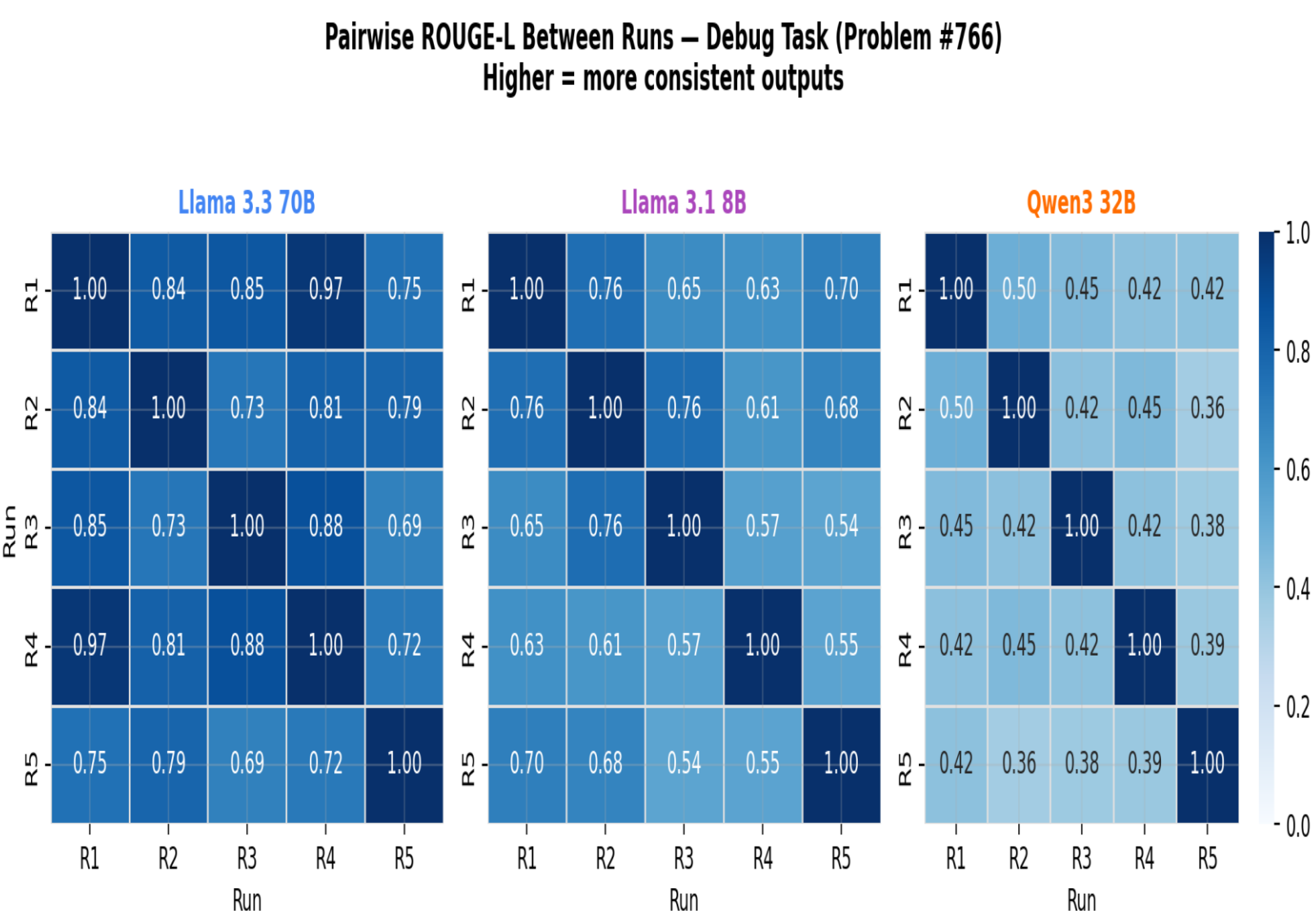


Figure 3. Consistency heatmap - models × task types.

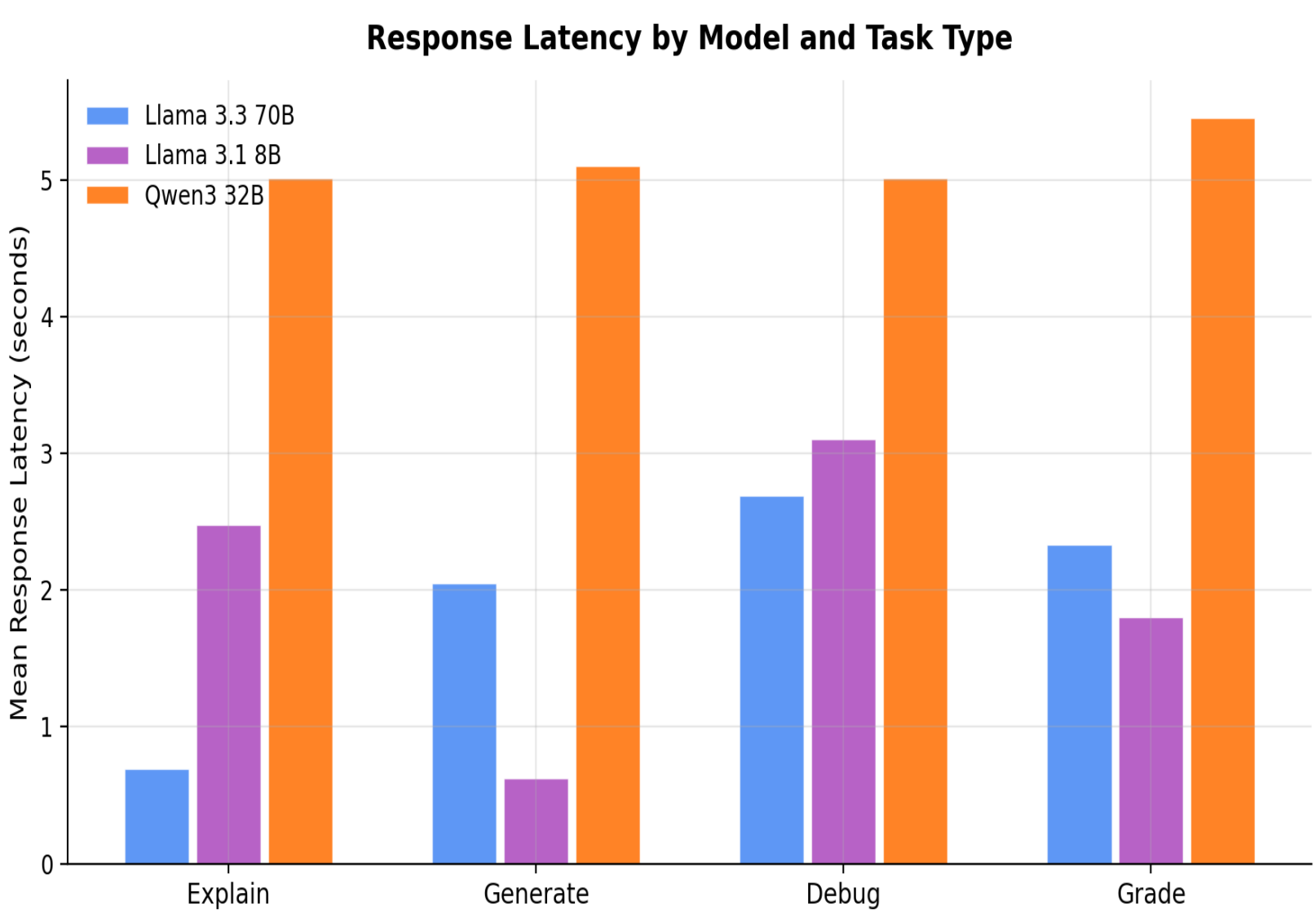


Figure 4. Latency by model across repeated runs.

Conclusions

Key takeaway

Evaluation must move beyond single-run accuracy to capture repeatability, stability, and cost.

Key contribution

A reproducible scaffold that evaluates LLMs across multiple runs capturing variability, stability limits, and cost behavior so results can be compared and replicated.

Practical value

Instructors need evidence of reliability, not just usefulness. This scaffold provides that in a standardised, reusable form.

Additional Information/Next Steps

- Expand to more benchmarks and authentic student tasks.
- Add prompt-robustness and rubric-sensitivity analysis.
- Build practical guidance for instructors choosing models.
- Support fair, reusable comparisons across future studies.

Long-term vision

This scaffold can be distilled into a lightweight, open-source model deployable directly to students at zero cost, giving them reliable, trustworthy AI feedback and a genuine reason to trust LLMs in their learning.

References

- [1] Pitts, G., Hridi, A. P., & Lekshmi-Narayanan, A. (2025). A survey of LLM-based applications in programming education: Balancing automation and human oversight. In Proceedings of the 4th Workshop on Bridging Human–Computer Interaction and Natural Language Processing (HCI+NLP), ACL 2025. <https://aclanthology.org/2025.hcinlp-1.21.pdf>
- [2] Alvarado Gonzalez, M. A., Bruno Hernandez, M., Peñaloza Perez, M. A., Lopez Orozco, B., Cruz Soto, J. T., & Malagon, S. (2025). Do repetitions matter? Strengthening reliability in LLM evaluations. arXiv preprint arXiv:2509.24086. <https://arxiv.org/abs/2509.24086>